Zyphroa – E-Commerce Platform

By :- DARSHAN M S

Employee ID :- 0060P7

Quality Engineer-Mainframe -2025

Email ID:- M.S.DARSHAN@ibm.com

1. Introduction.

Zyphroa is a fully functional front-end e-commerce platform developed using **HTML, CSS, and JavaScript**. The application simulates a real-world online shopping system where users can register, log in, browse products, add items to the cart, complete checkout, make payments, and view order history. All data is managed using the browser's **Local Storage**, eliminating the need for a backend server while still maintaining persistent user and order information.

The platform is structured to replicate the workflow of leading e-commerce websites such as Flipkart and Amazon, focusing on user authentication, product management, cart functionality, checkout validation, and order tracking. The system includes multiple product categories such as electronics, accessories, and home appliances, each displayed in a responsive grid layout with detailed product information including price, rating, description, and images.

To ensure the reliability, functionality, and stability of the application, automated testing was implemented using **Java, Maven, and Selenium WebDriver**. Apache Maven is used as the build and dependency management tool, allowing smooth execution of test suites and project compilation. Selenium WebDriver is used to automate browser actions such as login validation, cart operations, checkout process, navigation

flow, and session management. This ensures that each feature of the Zyphroa platform works as expected under different test scenarios.

For advanced reporting and better visualization of test execution results, the Allure Report framework is integrated into the project. Allure generates detailed, interactive test reports that include passed, failed, and skipped test cases, execution steps, logs, and overall test statistics. This enhances debugging capabilities and provides a professional-level reporting structure suitable for academic submissions and industry standards.

Overall, Zyphroa not only demonstrates strong front-end development capabilities but also showcases structured automation testing, build management, and professional reporting practices, making it a complete end-to-end e-commerce and QA automation project.

1.1 Objective of the Project.

The objective of this project is to design and develop a fully functional e-commerce system that simulates a real-world online shopping platform. The application includes user registration, login authentication, product browsing, cart management, checkout processing, payment simulation, and order history tracking. The project aims to ensure that all modules work smoothly and efficiently through structured testing.

Automation testing is implemented using Selenium WebDriver to validate different user flows such as login, adding products to cart, placing orders, and logout functionality. The project uses Apache Maven for build management, dependency handling, and smooth execution of test cases. Additionally, Allure Report is integrated to generate detailed and professional test execution reports.

Overall, the objective is to combine web development with automation testing and reporting tools to demonstrate a complete software testing lifecycle in a practical and structured manner.

1.2 Technologies Used

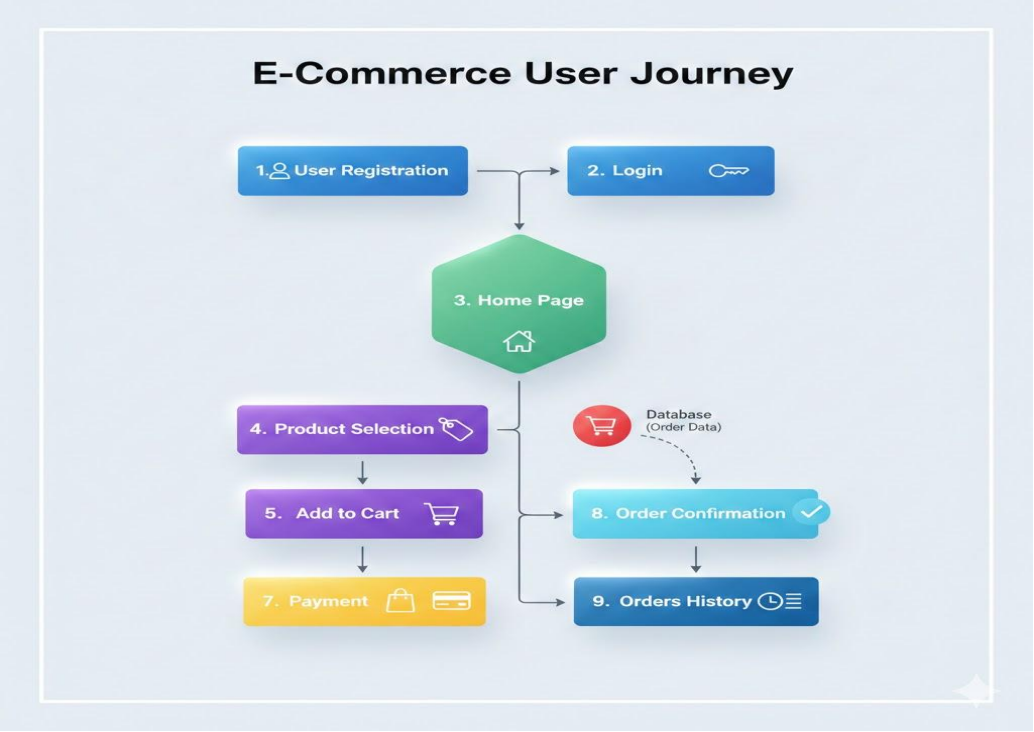
- Frontend: HTML5, CSS3, Vanilla JavaScript
- Storage: Local Storage
- Automation: Java, Selenium WebDriver, Maven
- Reporting: Allure Framework

Framework Design Pattern

The automation suite for Zyphroa utilizes the **Page Object Model (POM)** design pattern.

- **Maintenance:** By separating the page-specific locators (like Login fields or Add to Cart buttons) from the actual test scripts, the framework becomes highly maintainable.
- **Reusability:** Common actions like login() or MapsToCart() are stored in a Base Class, reducing code duplication across the 80 test cases.

1.3 Application Flow



1.4 Product Categories

- **Electronics**: Smart TVs, Smartphones, Headphones
- **Accessories**: Smart Watches, Earbuds, Laptop Bags
- **Home Appliances**: Refrigerator, Washing Machine, Air Conditioner.

2. Testing Strategy

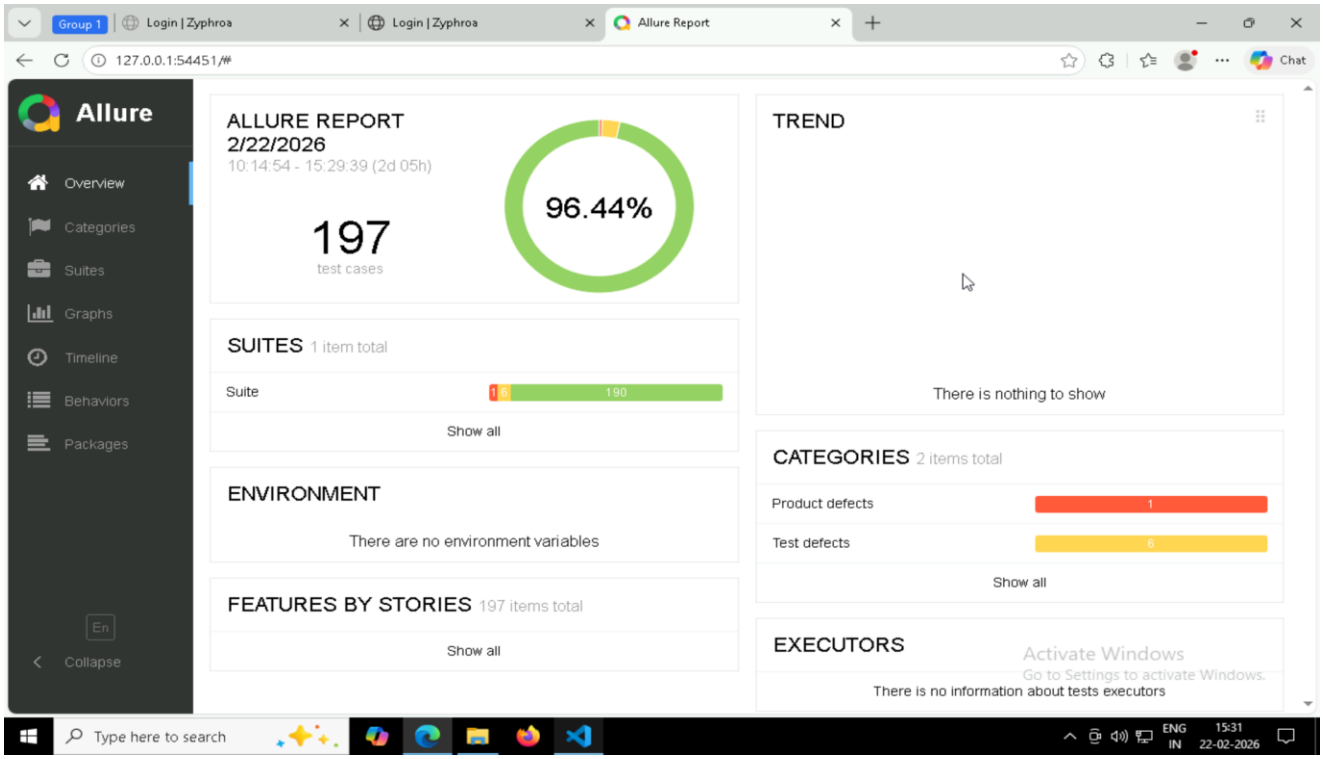
- **Functional Testing** – Validates core features.
- **UI Testing** – Ensures proper layout and element visibility.
- **Validation Testing** – Checks form validations.
- **Security Testing** – Prevents unauthorized access.
- **Performance Testing** – Verifies load time.
- **Session Testing** – Ensures login persistence.

Build Case

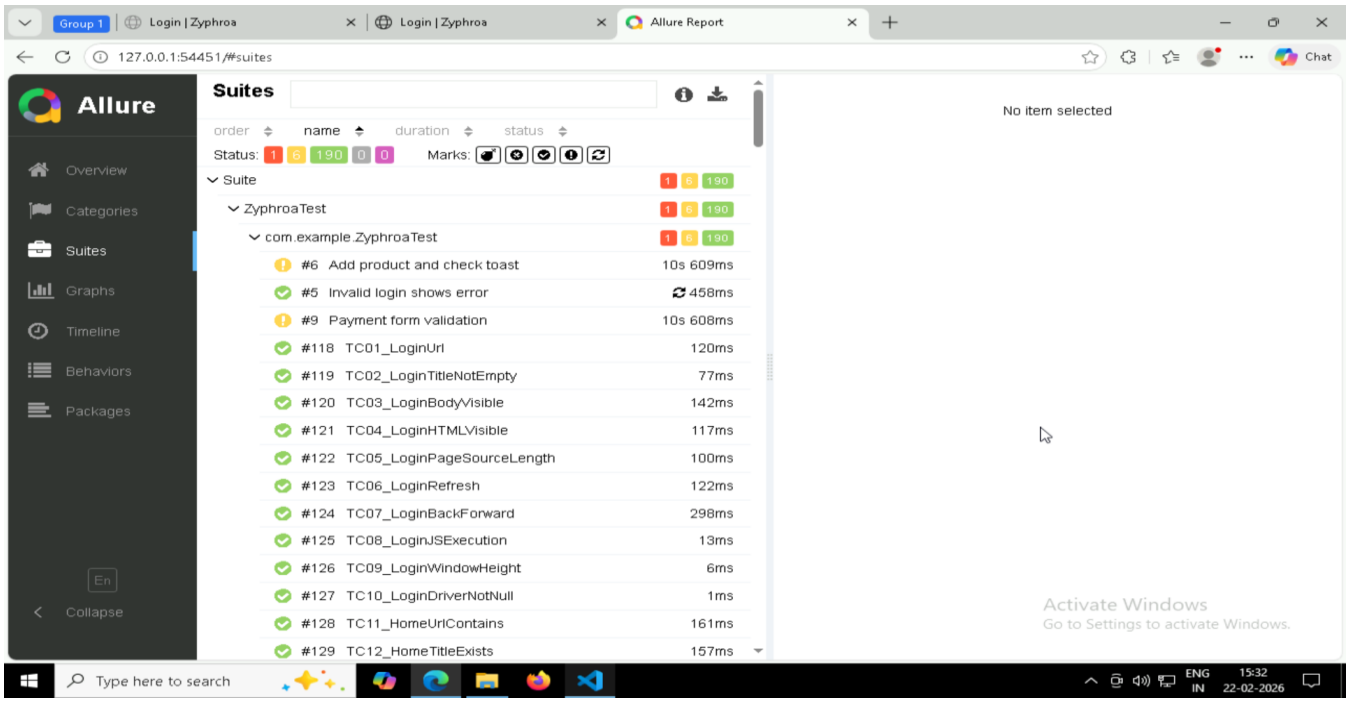
```
[INFO] Tests run: 80, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 20.42 s -- in TestSuite
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 80, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  25.515 s
[INFO] Finished at: 2026-02-22T17:51:58+05:30
[INFO] ------------------------------------------------------------------------
```

3. Results.

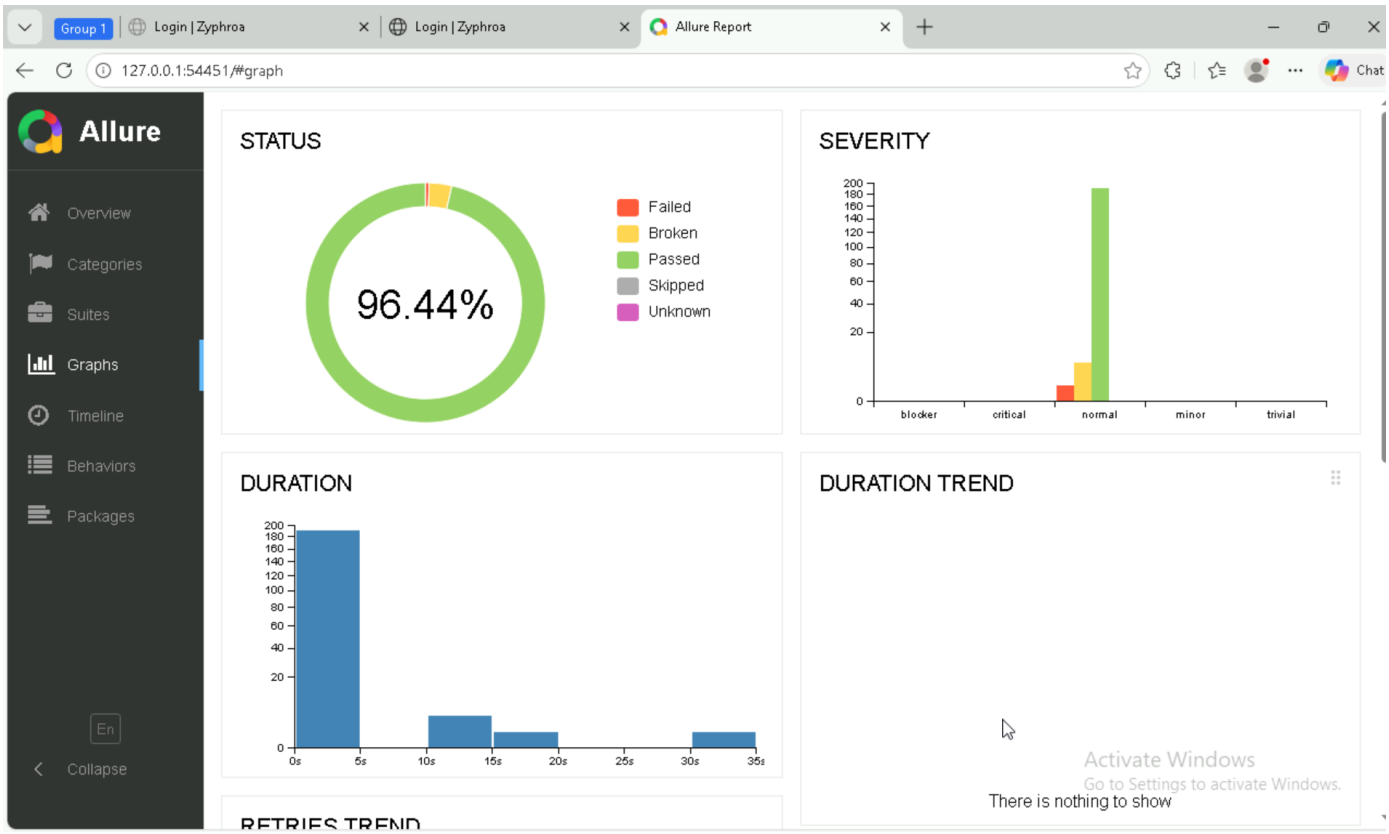
Overview:-



Suites:-



Graphs: -


Group 1
Login | Zyphroa
Login | Zyphroa
Allure Report
127.0.0.1:54451/#graph
Allure
Overview
Categories
Suites
Graphs
Timeline
Behaviors
Packages
En
Collapse
STATUS
96.44%
Failed
Broken
Passed
Skipped
Unknown
SEVERITY
blocker
critical
normal
minor
trivial
DURATION
DURATION TREND
Activate Windows
Go to Settings to activate Windows.
There is nothing to show

Zyphroa E-Commerce Platform - Test Cases Report

TC ID	Module	Test Case Description	Preconditions	Test Steps	Expected Result	Priority
TC_01	Login	Verify login page URL	Browser launched	Open login URL	URL ends with index.html	High
TC_02	Login	Verify login page title	Login page open	Get page title	Title not empty	Medium
TC_03	Login	Verify email field visible	Login page open	Locate email input	Email field displayed	High
TC_04	Login	Verify password field visible	Login page open	Locate password field	Password field displayed	High
TC_05	Login	Verify login button visible	Login page open	Locate login button	Button displayed	High
TC_06	Login	Verify login with valid credentials	Valid user exists	Enter valid credentials & login	Redirect to home page	Critical
TC_07	Login	Verify login with invalid credentials	Login page open	Enter invalid credentials	Error message displayed	High
TC_08	Login	Verify empty email validation	Login page open	Leave email empty	Validation message shown	High
TC_09	Login	Verify empty password validation	Login page open	Leave password empty	Validation message shown	High
TC_10	Login	Verify login page refresh	Login page open	Refresh page	Page reloads successfully	Medium
TC_11	Home	Verify home page URL	Logged in	Open home page	URL contains home.html	High
TC_12	Home	Verify product cards displayed	Home loaded	Inspect product section	Products visible	High
TC_13	Home	Verify product name visible	Home loaded	Check product title	Product name displayed	Medium
TC_14	Home	Verify product price visible	Home loaded	Check product price	Price displayed	High
TC_15	Home	Verify Add to Cart button	Home loaded	Locate add button	Button visible	High
TC_16	Home	Verify Add to Cart functionality	Product available	Click add button	Item added to cart	Critical
TC_17	Home	Verify search functionality	Search bar present	Search product	Matching results shown	High
TC_18	Home	Verify scroll down	Home loaded	Scroll page	Scroll works	Medium
TC_19	Home	Verify scroll up	Home loaded	Scroll to top	Scroll works	Medium
TC_20	Home	Verify home page refresh	Home loaded	Refresh	Stable reload	Medium
TC_21	Home	Verify navigation to cart	Home loaded	Click cart link	Cart page opens	High
TC_22	Home	Verify back navigation	Navigated away	Click back	Returns home	Medium
TC_23	Home	Verify forward navigation	After back	Click forward	Navigates correctly	Medium
TC_24	Home	Verify ready state complete	Home loaded	Check document.readyState	complete	High
TC_25	Home	Verify page source not blank	Home loaded	Get page source	Length > 200	Medium
TC_26	Cart	Verify cart page URL	Logged in	Open cart page	URL contains cart.html	High
TC_27	Cart	Verify cart items displayed	Items added	Open cart	Items visible	Critical
TC_28	Cart	Verify product name in cart	Items added	Check name	Correct name shown	High
TC_29	Cart	Verify product price in cart	Items added	Check price	Correct price shown	High
TC_30	Cart	Verify quantity increase	Item in cart	Increase quantity	Quantity updated	High
TC_31	Cart	Verify quantity decrease	Item in cart	Decrease quantity	Quantity updated	High
TC_32	Cart	Verify remove item	Item in cart	Click remove	Item removed	Critical
TC_33	Cart	Verify empty cart message	No items	Open cart	Cart is empty shown	Medium
TC_34	Cart	Verify total price calculation	Items added	Check total	Correct total displayed	Critical

TC ID	Module	Test Case Description	Preconditions	Test Steps	Expected Result	Priority
TC_35	Cart	Verify checkout button visible	Item in cart	Locate checkout	Button visible	High
TC_36	Cart	Verify checkout navigation	Item in cart	Click checkout	Checkout opens	Critical
TC_37	Cart	Verify cart refresh stability	Cart loaded	Refresh	Data persists	High
TC_38	Cart	Verify cart scroll	Cart loaded	Scroll page	Scroll works	Medium
TC_39	Cart	Verify cart ready state	Cart loaded	Check readyState	complete	High
TC_40	Cart	Verify back navigation from cart	Cart open	Click back	Works properly	Medium
TC_41	Checkout	Verify checkout URL	Logged in	Open checkout	URL correct	High
TC_42	Checkout	Verify name field visible	Checkout page	Locate name field	Field visible	High
TC_43	Checkout	Verify address field visible	Checkout page	Locate address field	Field visible	High
TC_44	Checkout	Verify payment options visible	Checkout page	Inspect payment section	Options displayed	High
TC_45	Checkout	Verify empty form validation	Checkout open	Submit empty form	Validation shown	High
TC_46	Checkout	Verify valid form submission	Valid data	Submit form	Order placed	Critical
TC_47	Checkout	Verify cancel button	Checkout open	Click cancel	Returns to cart	Medium
TC_48	Checkout	Verify checkout refresh	Checkout open	Refresh	Stable reload	Medium
TC_49	Checkout	Verify total amount display	Checkout open	Check total	Correct amount	High
TC_50	Checkout	Verify order confirmation message	Order placed	Observe confirmation	Success message shown	Critical
TC_51	Checkout	Verify required fields marked	Checkout open	Inspect fields	Required fields marked	Medium
TC_52	Checkout	Verify back navigation	Checkout open	Click back	Returns cart	Medium
TC_53	Checkout	Verify ready state	Checkout open	Check readyState	complete	High
TC_54	Checkout	Verify scroll	Checkout open	Scroll page	Works	Medium
TC_55	Checkout	Verify direct URL access without login	Not logged in	Open checkout URL	Redirect login	Critical
TC_56	Orders	Verify orders page URL	Logged in	Open orders	URL correct	High
TC_57	Orders	Verify order history displayed	Order placed	Open orders	Orders listed	Critical
TC_58	Orders	Verify order ID visible	Order placed	Inspect order	ID visible	High
TC_59	Orders	Verify order date visible	Order placed	Inspect date	Date shown	Medium
TC_60	Orders	Verify order total visible	Order placed	Inspect total	Correct amount	High
TC_61	Orders	Verify no orders message	No orders	Open orders	No orders shown	Medium
TC_62	Orders	Verify orders refresh	Orders open	Refresh	Stable	Medium
TC_63	Orders	Verify scroll	Orders open	Scroll page	Works	Low
TC_64	Orders	Verify ready state	Orders open	Check readyState	complete	High
TC_65	Orders	Verify back navigation	Orders open	Click back	Works	Medium
TC_66	Navigation	Verify browser back button	Any page	Click back	Navigates correctly	Medium
TC_67	Navigation	Verify browser forward button	After back	Click forward	Works	Medium
TC_68	Navigation	Verify multiple page navigation	Logged in	Navigate across pages	No crash	High
TC_69	Navigation	Verify logout functionality	Logged in	Click logout	Redirect login	Critical
TC_70	Session	Verify session persistence after refresh	Logged in	Refresh page	Still logged in	High
TC_71	Session	Verify localStorage contains login data	Logged in	Check localStorage	Data exists	High

TC ID	Module	Test Case Description	Preconditions	Test Steps	Expected Result	Priority
TC_72	Session	Verify session expired redirect	Logged out	Access home URL	Redirect login	High
TC_73	Navigation	Verify window resize stability	Any page	Resize window	Layout stable	Medium
TC_74	Performance	Verify page load time under 5 sec	Any page	Measure load time	Less than 5 sec	Medium
TC_75	Console	Verify no major console errors	Any page	Inspect console	No critical errors	Medium
TC_76	Security	Verify direct cart access without login	Not logged in	Open cart URL	Redirect login	High
TC_77	Security	Verify direct orders access without login	Not logged in	Open orders URL	Redirect login	High
TC_78	Stability	Verify multiple refresh stability	Any page	Refresh 3 times	No crash	Low
TC_79	System	Verify WebDriver session active	Driver initialized	Check driver object	Not null	Low
TC_80	System	Verify document ready state on all pages	Any page	Check readyState	complete	High

Conclusion

In conclusion, the Zyphroa E-commerce Platform successfully demonstrates the development of a complete online shopping system using front-end technologies along with structured automation testing practices. The project effectively integrates core web development concepts with automated validation to ensure functional accuracy and system stability. By using Selenium WebDriver for automation, Apache Maven for build management, and Allure Report for advanced reporting, the project follows industry-standard testing methodologies.

This project not only enhances practical knowledge of e-commerce workflows but also strengthens understanding of automation frameworks, test execution strategies, and professional reporting. Overall, Zyphroa represents a complete end-to-end implementation of development and quality assurance processes in a structured and efficient manner.